

Precalc Lesson 1.3 Homework

Name _____

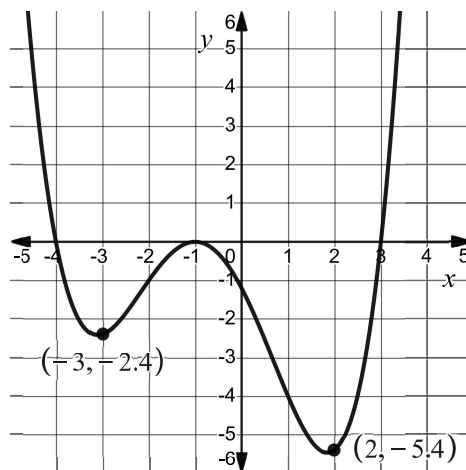
1. Let $k(x) = -(x + 1)^2 + 11$. Find the average rate of change of $k(x)$ between $x = -2$ and $x = 4$.

2. The temperature of a mug of hot chocolate, t minutes after it is prepared is given by the function $H(t)$ where H is measured in degrees Fahrenheit. Selected values of $H(t)$ are given below.

t	0	2	5	10	13	20
$H(t)$	170	137	107	83	77	72

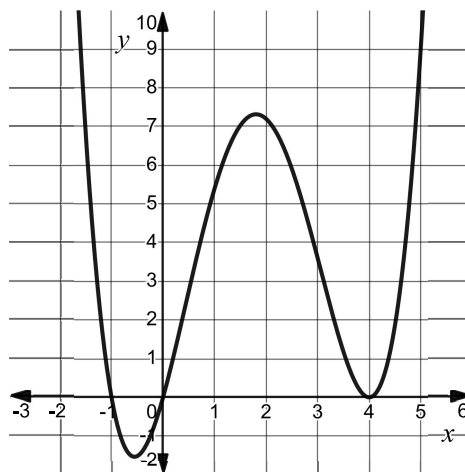
Find the average rate of change of the temperature of the hot chocolate between $t = 0$ and $t = 20$. Use proper units in your answer.

3. The graph of $y = h(x)$ is shown. Find the average rate of change of h on $[-3, 2]$.

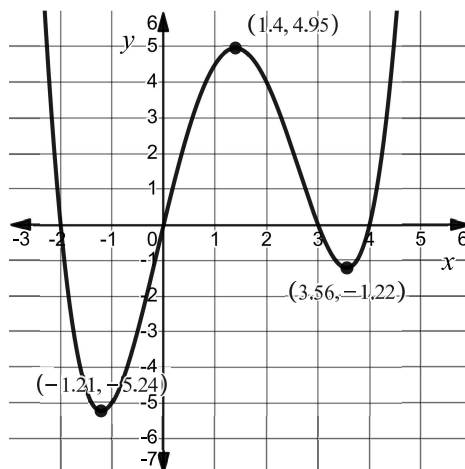


4. The graph of $y = f(x)$ is shown. Order the following from least to greatest.

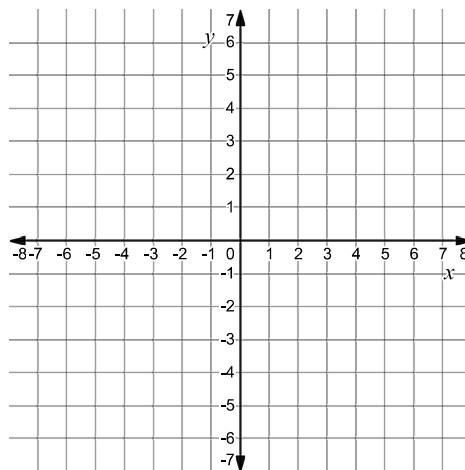
- A. Average rate of change of f on $[0, 4]$.
- B. Average rate of change of f on $[0, 2]$.
- C. Average rate of change of f on $[2, 4]$.
- D. Average rate of change of f on $[2, 5]$.



5. The graph of $y = f(x)$ is shown. Identify the intervals on which f is increasing, decreasing, and constant.



6. Sketch the graph of a function that is increasing for $x \leq -5$, constant for $-5 \leq x \leq 0$, increasing for $0 \leq x \leq 2$, and decreasing for $x \geq 2$.



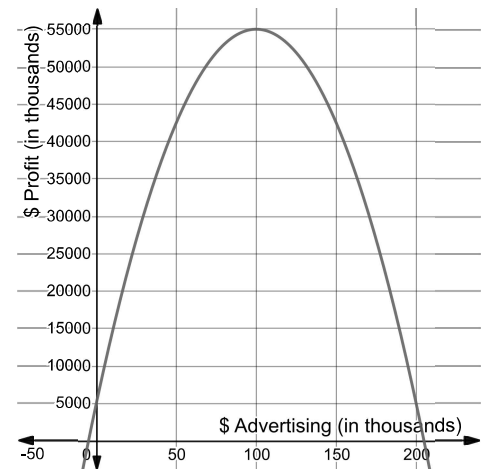
7. Information about a function $f(x)$ is given in the table.

x	$-\infty < x \leq -3$	$x = -3$	$-3 \leq x \leq 5$	$x = 5$	$5 \leq x < \infty$
$f(x)$	Decreasing	5	Increasing	11	Increasing

- a. For which value(s) of x , if any, does $f(x)$ have a relative (local) minimum?
- b. For which value(s) of x , if any, does $f(x)$ have a relative (local) maximum?
8. The value of stock in Chelsea's Chocolate Company increased between 2017 and 2019, then decreased until June of 2020, then started increasing again. Which of the following conclusions is true?
- A) The value of the stock was at its absolute highest in 2019.
- B) The value of the stock reached a relative maximum in 2019.
- C) The value of the stock reached a relative minimum in 2017.
- D) The value of the stock was at its absolute lowest in June of 2020.

9. After doing some market research, Crunchy Cookie Inc. has found that their profit on cookies after spending x amount on advertising can be modeled by $P(x) = 5000 + 1000x - 5x^2$. Both x and $P(x)$ are measured in thousands of dollars.

- What is the most profit that Crunchy Cookie Inc. can make?
- How much would they have to spend on advertising to achieve this result?
- Is there ever a time when Crunchy Cookie Inc. will make no profit? Explain why or why not.



10. The graph shows the rate at which people enter an art museum, $a(t)$, measured in people per hour, where t is measured in hours after the museum opens.

- Identify the interval(s) on which a is increasing. What does this mean in the context of this problem?
- Identify the interval(s) on which a is constant. What does this mean in the context of this problem?
- Identify the maximum of a and interpret its meaning in the context of this problem.
- When is $a(t) = 0$? What does this mean in the context of this problem?

